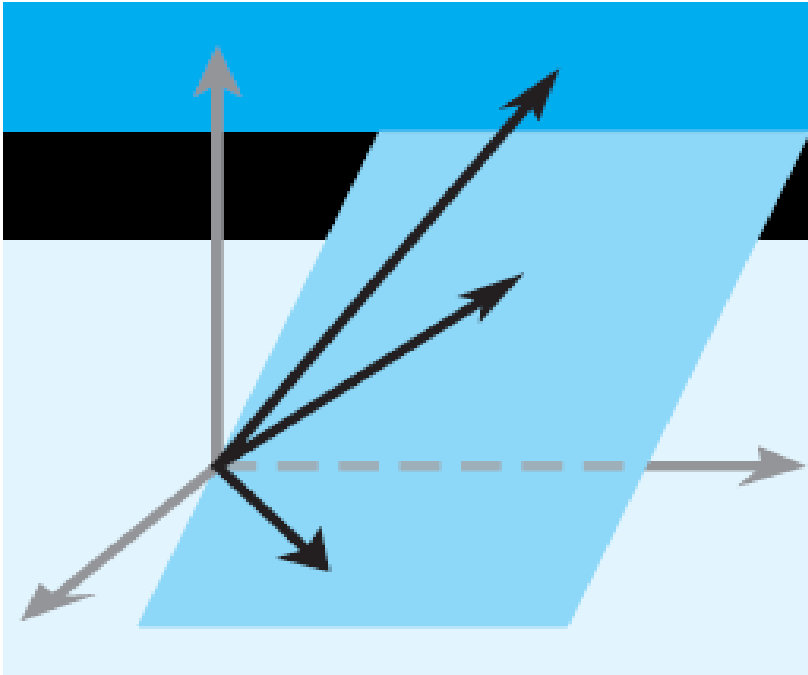


Elementary Linear Algebra



Chapter 4

Howard Anton

Copyright © 2014 by John Wiley & Sons, Inc.

All rights reserved.



Chapter 4

General Vector Spaces

- 4.1 Real Vector Spaces
- 4.2 Subspaces
- 4.3 Linear Independence
- 4.4 Coordinates and Basis
- 4.5 Dimension
- 4.6 Change of Basis
- 4.7 Row Space, Column Space, and Null Space
- 4.8 Rank, Nullity, and the Fundamental Matrix Spaces
- 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$
- 4.10 Properties of Matrix Transformations
- 4.11 Geometry of Matrix Operators on $\mathbb{R}^2$
- 4.12 Dynamical Systems and Markov Chains

Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

Section 4.11 Geometry of Matrix Operators on $\mathbb{R}^2$

DEFINITION 1 If V and W are vector spaces, and if f is a function with domain V and codomain W , then we say that f is a *transformation* from V to W or that f *maps* V to W , which we denote by writing

$$f: V \rightarrow W$$

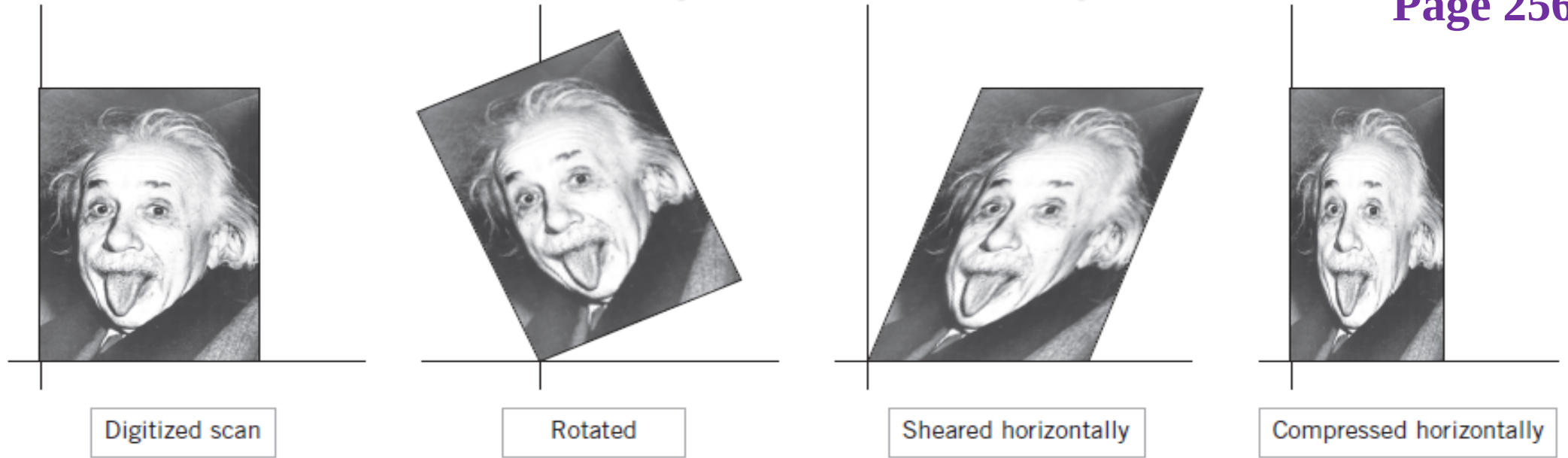
In the special case where $V = W$, the transformation is also called an *operator* on V .



Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

Section 4.11 Geometry of Matrix Operators on $\mathbb{R}^2$

Page 256



▲ Figure 4.11.1

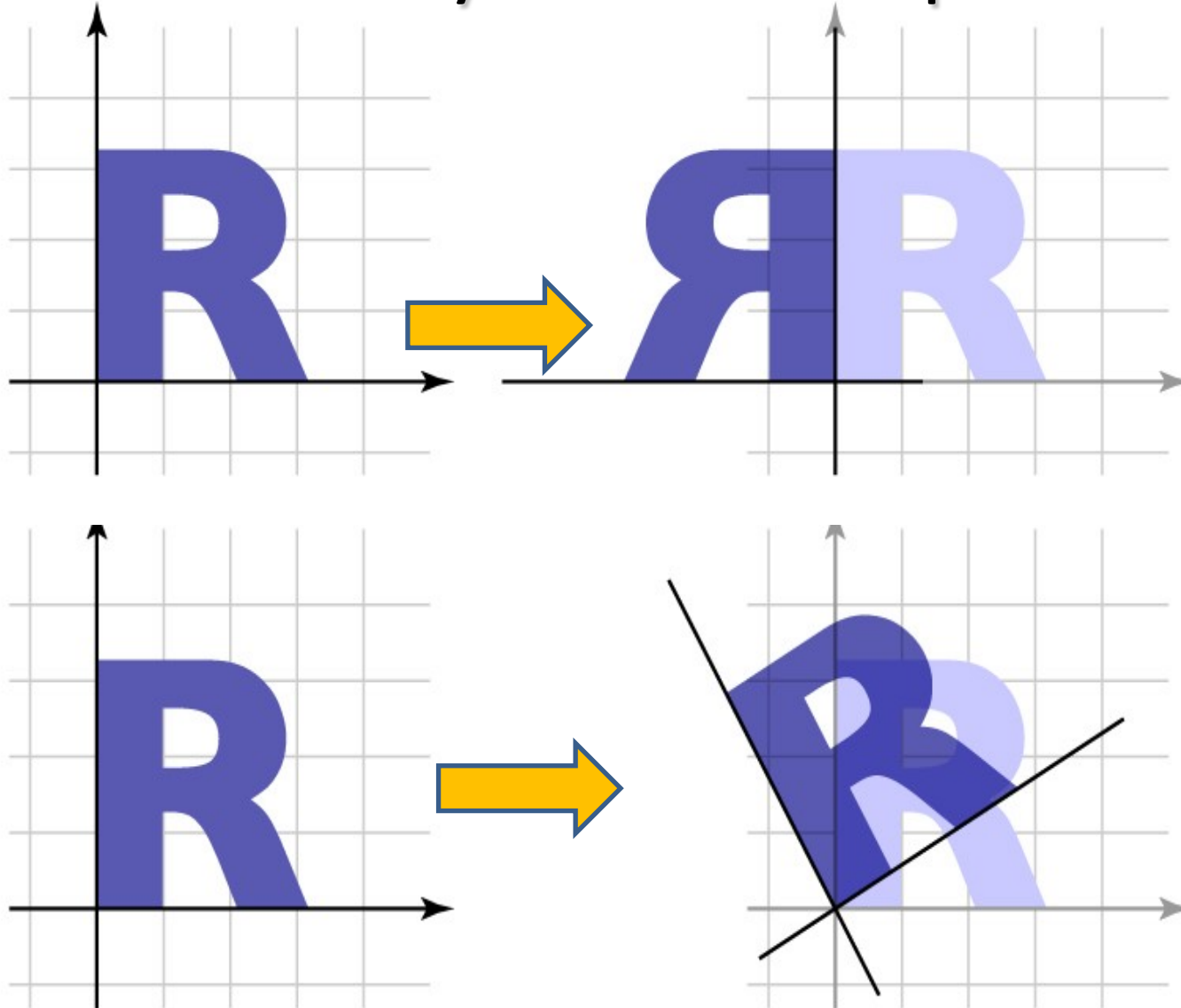
Chapter 8

平面上的線性變換，最基本的是下列的四種：旋轉、鏡射、伸縮、推移。

本文將介紹這四種線性變換，及其所對應表示的矩陣。

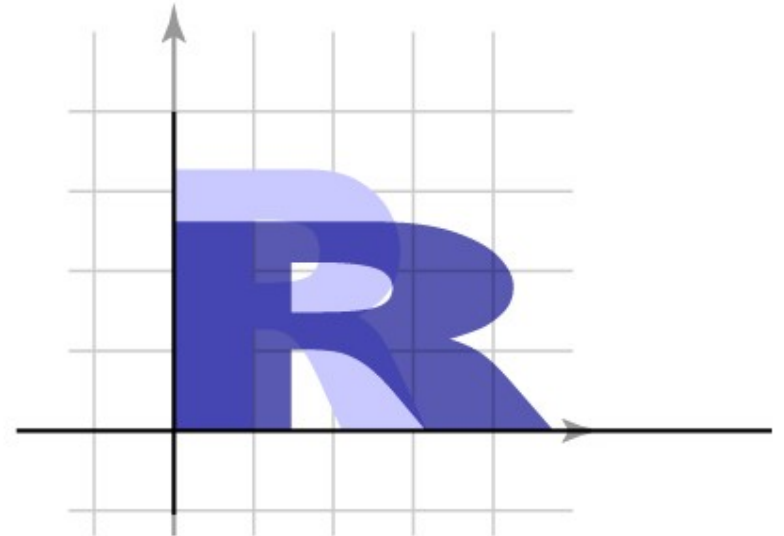
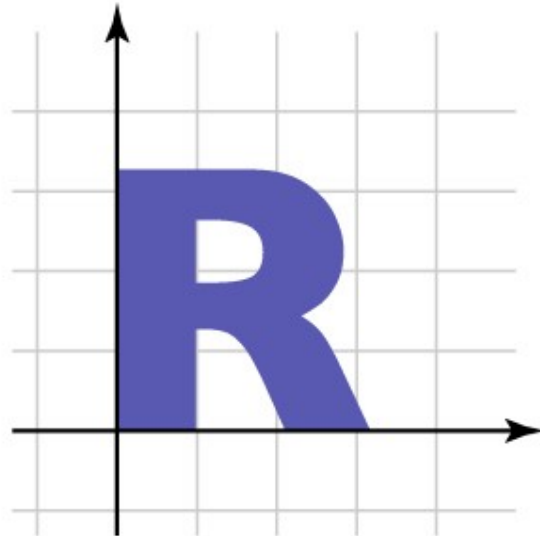
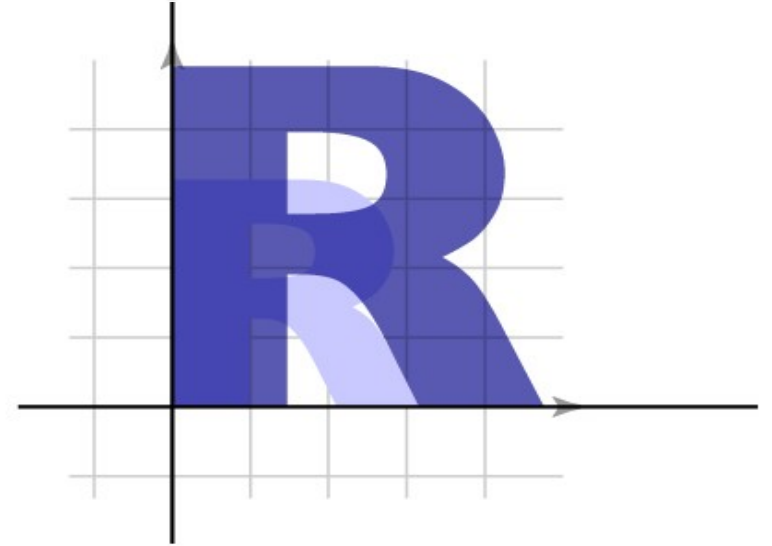
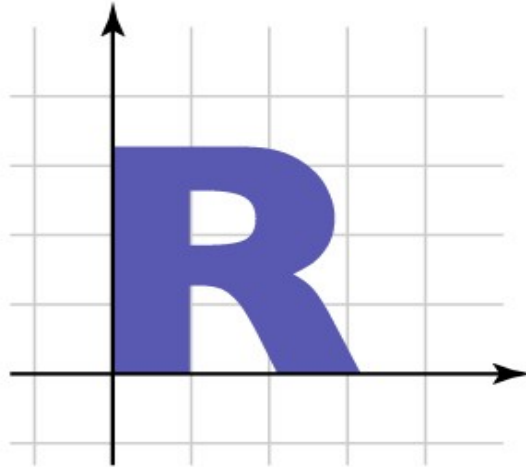
Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

Section 4.11 Geometry of Matrix Operators on $\mathbb{R}^2$



Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

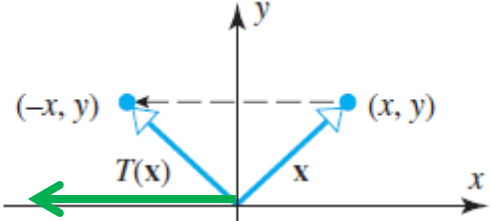
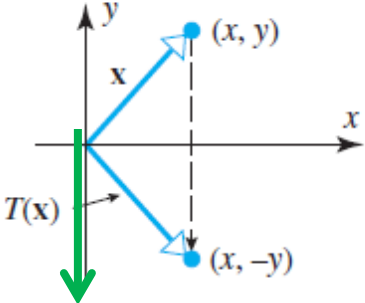
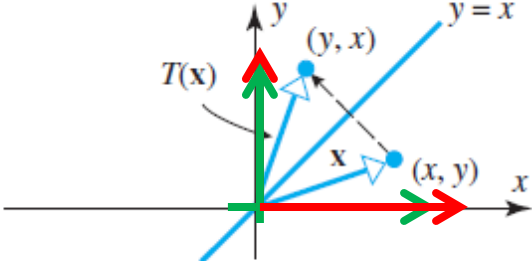
Section 4.11 Geometry of Matrix Operators on $\mathbb{R}^2$



Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

Section 4.11 Geometry of Matrix Operators on $\mathbb{R}^2$

Table 1

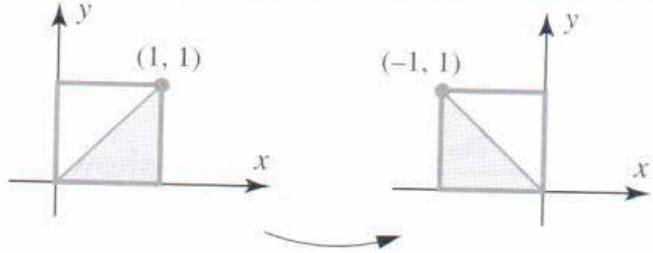
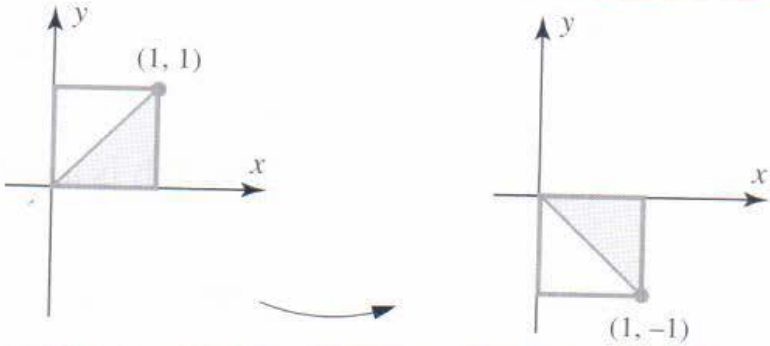
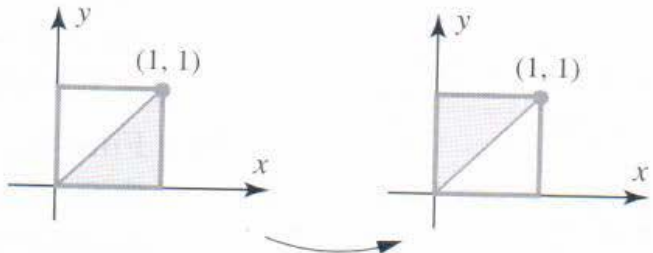
Operator	Illustration	Images of e_1 and e_2	Standard Matrix
Reflection about the y -axis $T(x, y) = (-x, y)$			
Reflection about the x -axis $T(x, y) = (x, -y)$			
Reflection about the line $y = x$ $T(x, y) = (y, x)$			

Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

Section 4.11 Geometry of Matrix Operators on $\mathbb{R}^2$

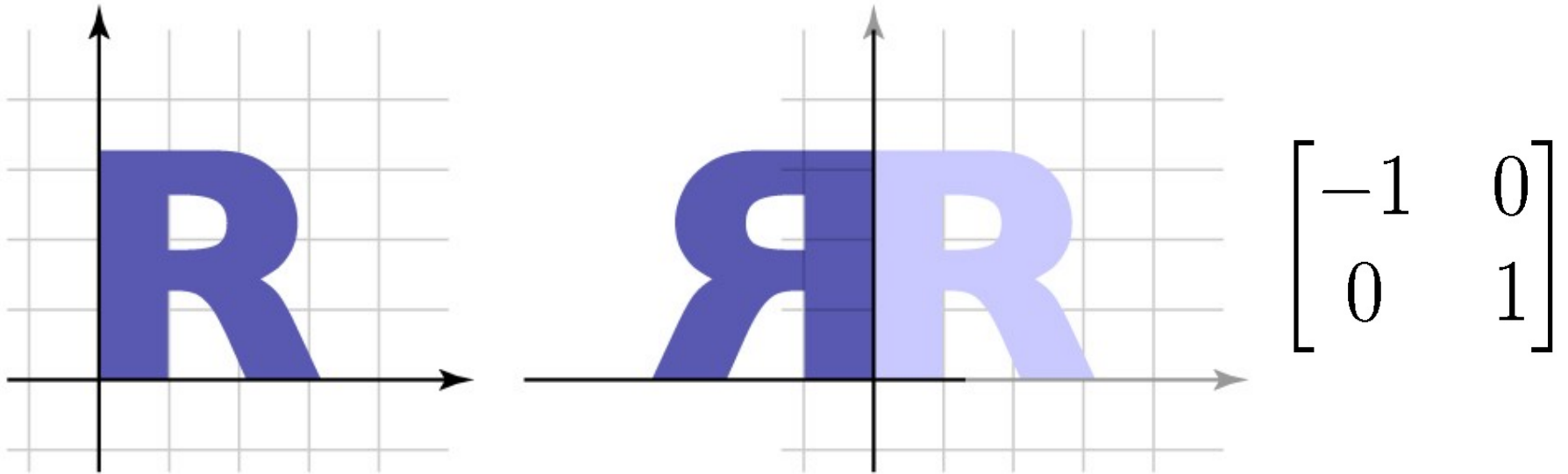
Table 1

Page 257

Operator	Standard Matrix	Effect on the Unit Square
Reflection about the y-axis	$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$	
Reflection about the x-axis	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	
Reflection about the line $y = x$	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	

Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

Section 4.11 Geometry of Matrix Operators on $\mathbb{R}^2$



Reflection about the y-axis

Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

Page 238

Table 2

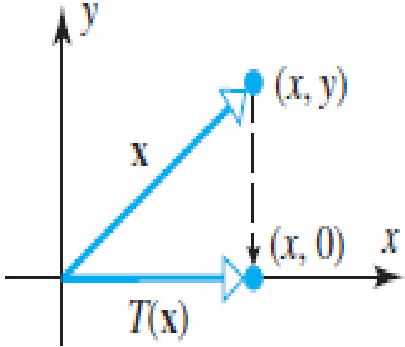
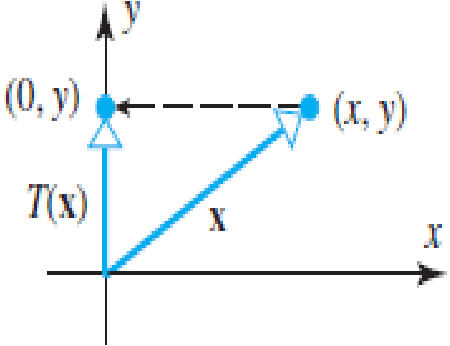
Operator	Illustration	e_1, e_2, e_3	Standard Matrix
Reflection about the xy-plane $T(x, y, z) = (x, y, -z)$		$T(e_1) = T(1, 0, 0) = (1, 0, 0)$ $T(e_2) = T(0, 1, 0) = (0, 1, 0)$ $T(e_3) = T(0, 0, 1) = (0, 0, -1)$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$
Reflection about the xz-plane $T(x, y, z) = (x, -y, z)$		$T(e_1) = T(1, 0, 0) = (1, 0, 0)$ $T(e_2) = T(0, 1, 0) = (0, -1, 0)$ $T(e_3) = T(0, 0, 1) = (0, 0, 1)$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
Reflection about the yz-plane $T(x, y, z) = (-x, y, z)$		$T(e_1) = T(1, 0, 0) = (-1, 0, 0)$ $T(e_2) = T(0, 1, 0) = (0, 1, 0)$ $T(e_3) = T(0, 0, 1) = (0, 0, 1)$	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

Section 4.11 Geometry of Matrix Operators on $\mathbb{R}^2$

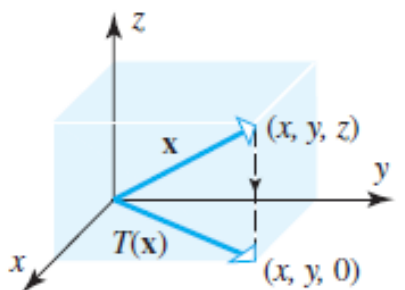
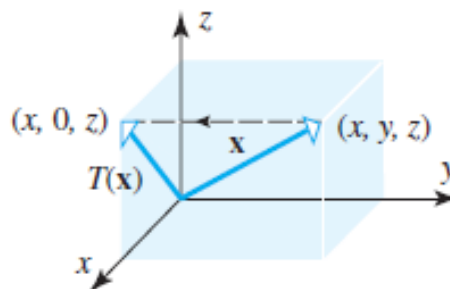
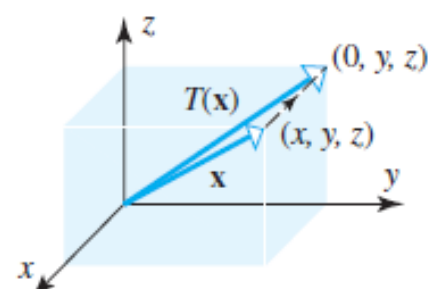
Table 3

Page 239

Operator	Illustration	Images of e_1 and e_2	Standard Matrix
Orthogonal projection on the x -axis $T(x, y) = (x, 0)$		$T(e_1) = T(1, 0) = (1, 0)$ $T(e_2) = T(0, 1) = (0, 0)$	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$
Orthogonal projection on the y -axis $T(x, y) = (0, y)$		$T(e_1) = T(1, 0) = (0, 0)$ $T(e_2) = T(0, 1) = (0, 1)$	$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

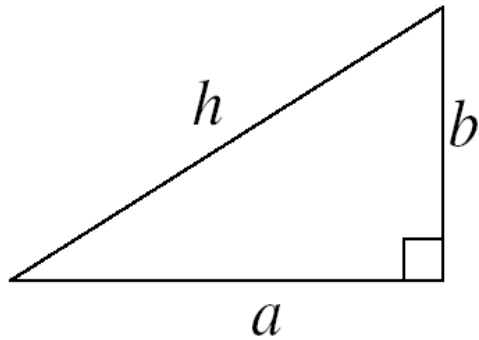
Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

Table 4

Operator	Illustration	Images of $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$	Standard Matrix
Orthogonal projection on the xy -plane $T(x, y, z) = (x, y, 0)$		$T(\mathbf{e}_1) = T(1, 0, 0) = (1, 0, 0)$ $T(\mathbf{e}_2) = T(0, 1, 0) = (0, 1, 0)$ $T(\mathbf{e}_3) = T(0, 0, 1) = (0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$
Orthogonal projection on the xz -plane $T(x, y, z) = (x, 0, z)$		$T(\mathbf{e}_1) = T(1, 0, 0) = (1, 0, 0)$ $T(\mathbf{e}_2) = T(0, 1, 0) = (0, 0, 0)$ $T(\mathbf{e}_3) = T(0, 0, 1) = (0, 0, 1)$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
Orthogonal projection on the yz -plane $T(x, y, z) = (0, y, z)$		$T(\mathbf{e}_1) = T(1, 0, 0) = (0, 0, 0)$ $T(\mathbf{e}_2) = T(0, 1, 0) = (0, 1, 0)$ $T(\mathbf{e}_3) = T(0, 0, 1) = (0, 0, 1)$	$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Section 4.9 Matrix Transformations from R^n to R^m

Review



Pythagoras's Theorem

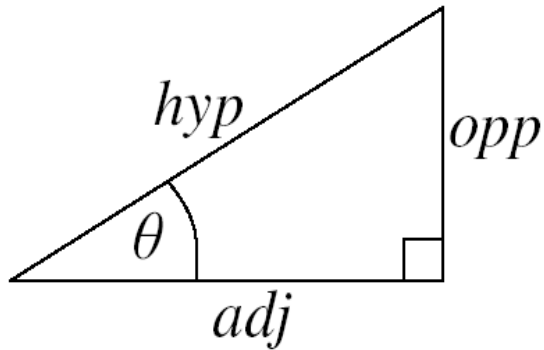
$$a^2 + b^2 = h^2$$

Trigonometric Ratios

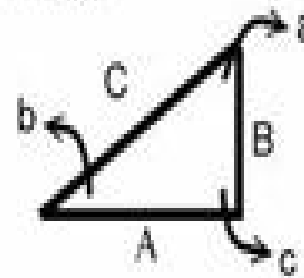
$$\sin(\theta) = \frac{\text{opp}}{\text{hyp}}$$

$$\cos(\theta) = \frac{\text{adj}}{\text{hyp}}$$

$$\tan(\theta) = \frac{\text{opp}}{\text{adj}}$$



直角三角形



A, B, C 為三角形三個邊
 a, b, c 為三角形三個角
其中 c 為直角(90度)：所以
 $a+b=90$

畢氏定理 $C^2 = A^2 + B^2$

正弦定理 ($\frac{\text{對邊}}{\text{斜邊}}$) $\sin a = \frac{A}{C}$, $\sin b = \frac{B}{C}$

餘弦定理 ($\frac{\text{鄰邊}}{\text{斜邊}}$) $\cos a = \frac{B}{C}$, $\cos b = \frac{A}{C}$

正切定理 ($\frac{\text{對邊}}{\text{鄰邊}}$) $\tan a = \frac{A}{B}$, $\tan b = \frac{B}{A}$

Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

Section 4.11 Geometry of Matrix Operators on $\mathbb{R}^2$

Rotation Operators

<https://www.geogebra.org/m/qjH94neE>

<https://www.geogebra.org/m/pnveFCjE>

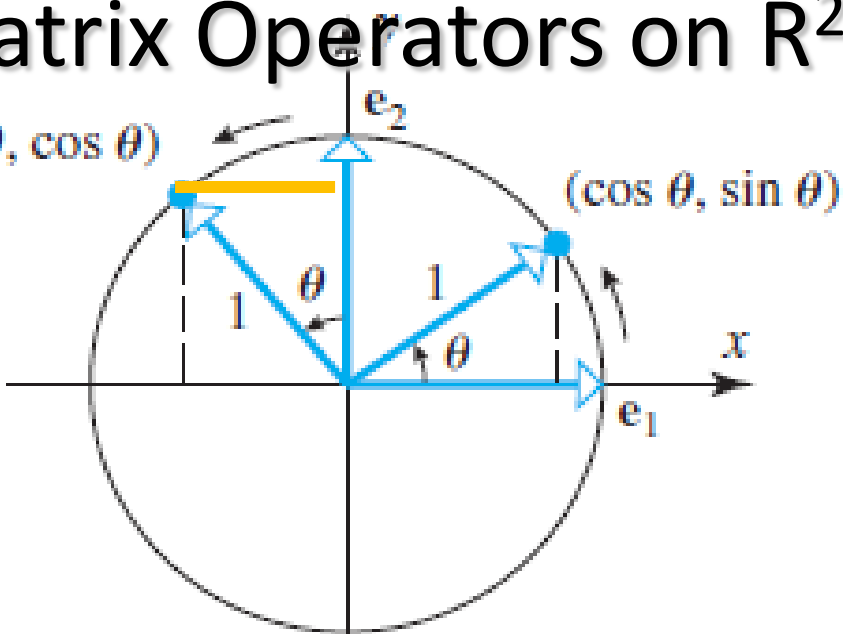


Table 5

Operator	Illustration	Rotation Equations	Standard Matrix
Rotation through an angle θ		$\begin{aligned}w_1 &= x \cos \theta - y \sin \theta \\w_2 &= x \sin \theta + y \cos \theta\end{aligned}$	$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$

Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

Section 4.11 Geometry of Matrix Operators on $\mathbb{R}^2$

► EXAMPLE 4 A Rotation Operator

Find the image of $\mathbf{x} = (1, 1)$ under a rotation of $\pi/6$ radians ($=30^\circ$) about the origin.

Solution It follows from (13) with $\theta = \pi/6$ that

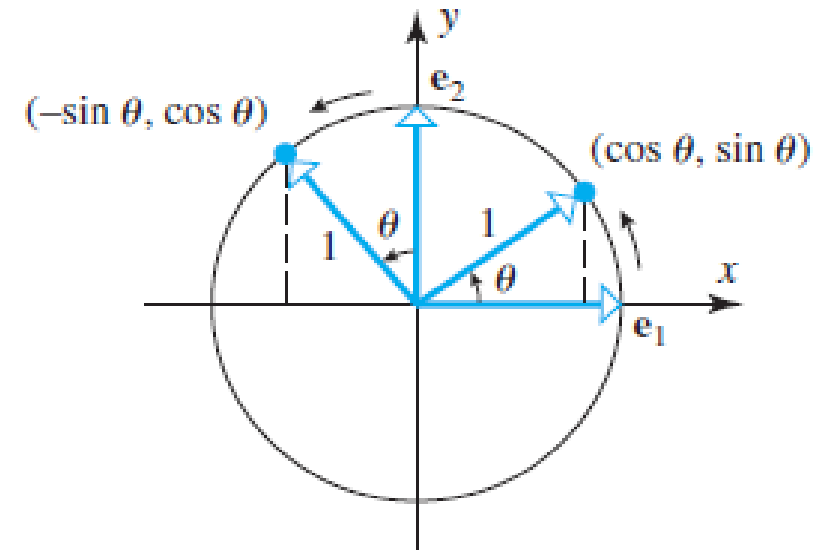
$$R_{\pi/6} \mathbf{x} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}-1}{2} \\ \frac{1+\sqrt{3}}{2} \end{bmatrix} \approx \begin{bmatrix} 0.37 \\ 1.37 \end{bmatrix}$$

$$R_{\pi/6} (1, 1) \approx (0.37, 1.37) .$$

or in comma-delimited notation,

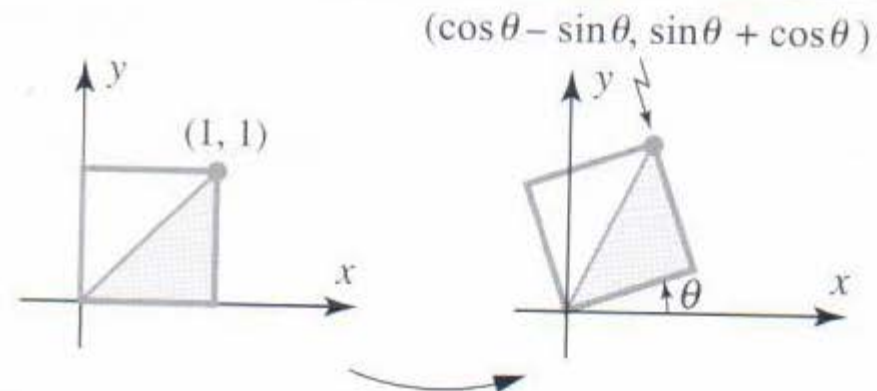
Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

Section 4.11 Geometry of Matrix Operators on $\mathbb{R}^2$



Counterclockwise
rotation through
an angle θ

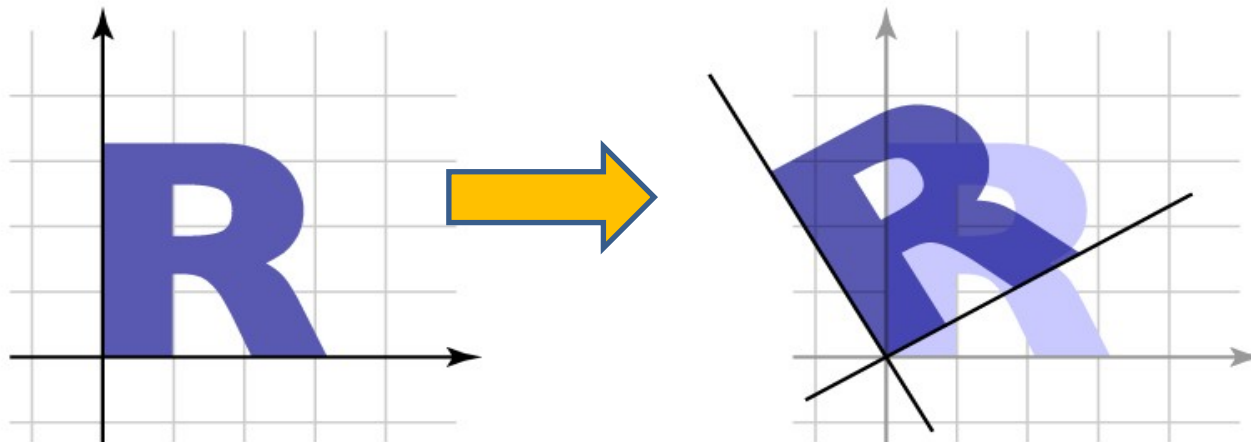
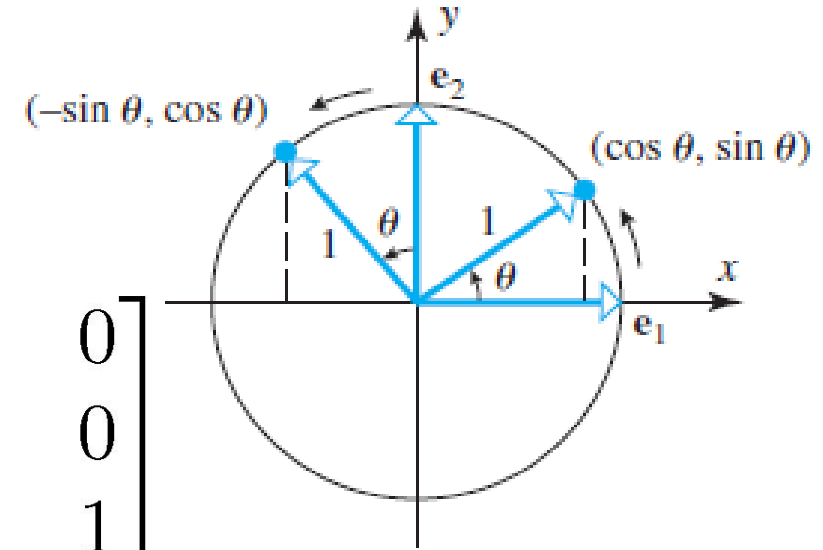
$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$



Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

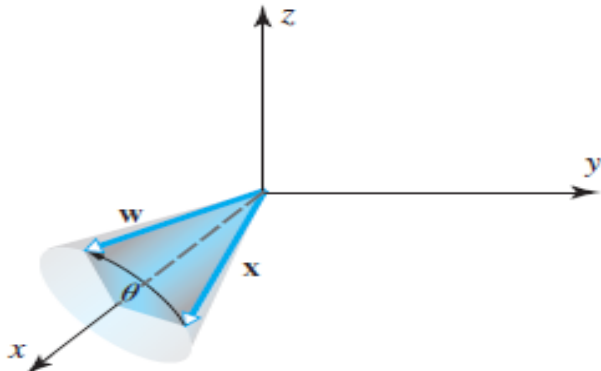
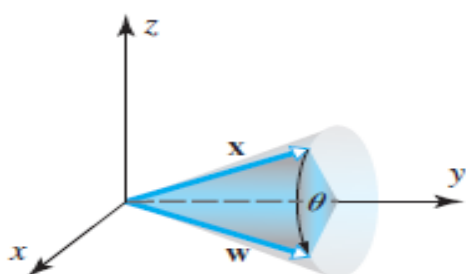
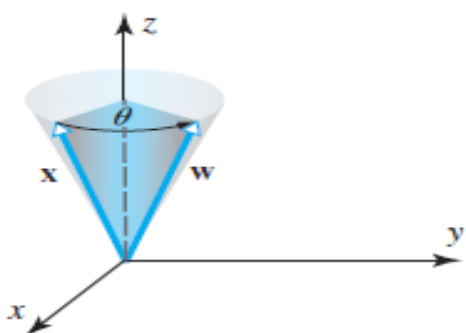
Rotation Operators

$$\begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

Table 6

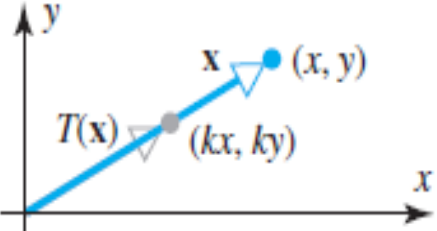
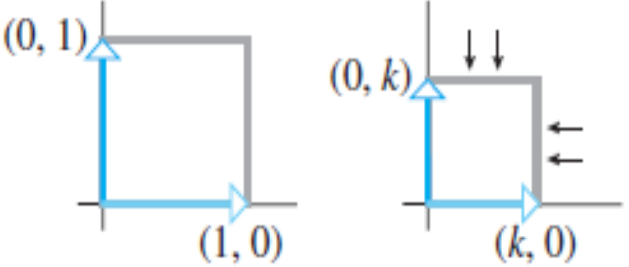
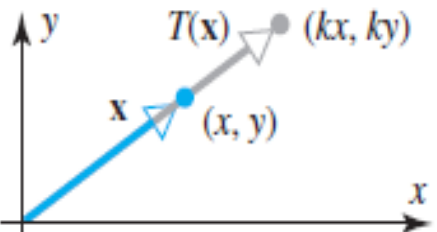
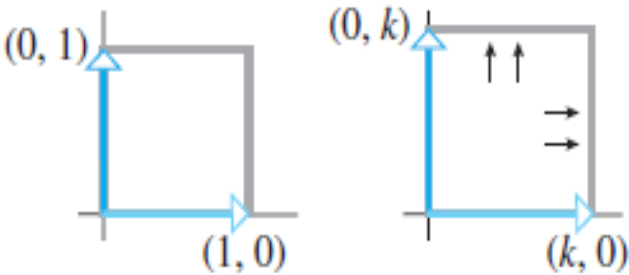
Operator	Illustration	Rotation Equations	Standard Matrix
Counterclockwise rotation about the positive x -axis through an angle θ		$w_1 = x$ $w_2 = y \cos \theta - z \sin \theta$ $w_3 = y \sin \theta + z \cos \theta$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$
Counterclockwise rotation about the positive y -axis through an angle θ		$w_1 = x \cos \theta + z \sin \theta$ $w_2 = y$ $w_3 = -x \sin \theta + z \cos \theta$	$\begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$
Counterclockwise rotation about the positive z -axis through an angle θ		$w_1 = x \cos \theta - y \sin \theta$ $w_2 = x \sin \theta + y \cos \theta$ $w_3 = z$	$\begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

Section 4.11 Geometry of Matrix Operators on $\mathbb{R}^2$

Dilations and Contractions

Table 7

Operator	Illustration $T(x, y) = (kx, ky)$	Effect on the Standard Basis	Standard Matrix
Contraction with factor k on $\mathbb{R}^2$ $(0 \leq k < 1)$			$\begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$
Dilation with factor k on $\mathbb{R}^2$ $(k > 1)$			

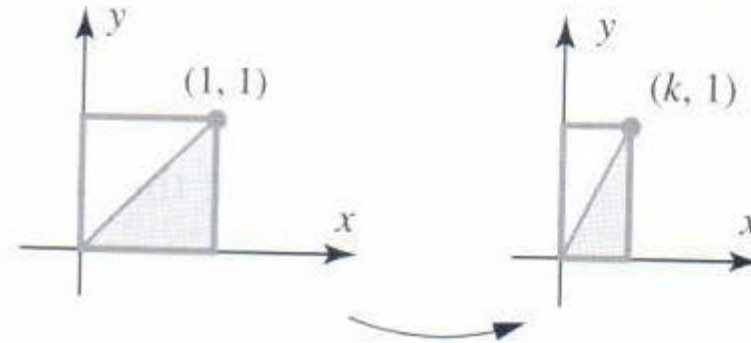
Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

Section 4.11 Geometry of Matrix Operators on $\mathbb{R}^2$

Compression in the
 x -direction by a
factor of k

$(0 < k < 1)$

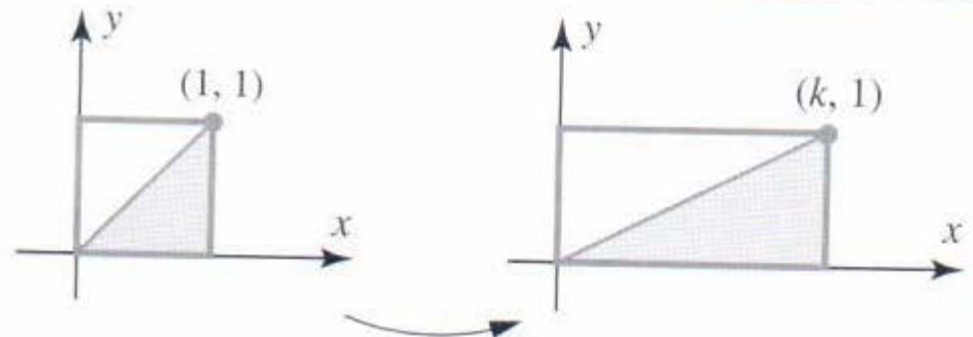
$$\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$$



Expansion in the
 x -direction by a
factor of k

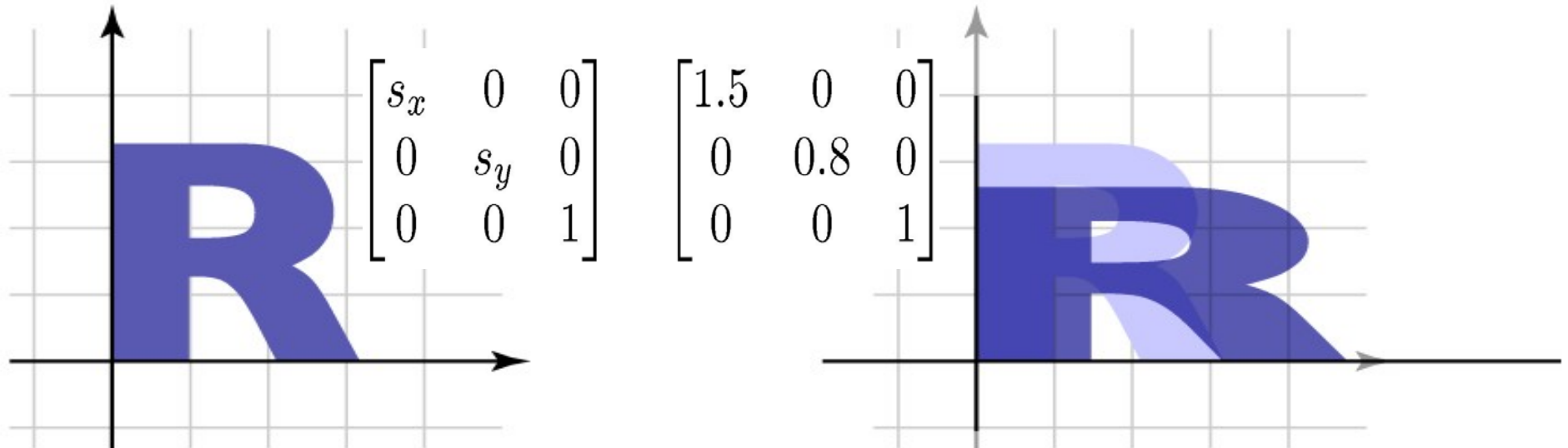
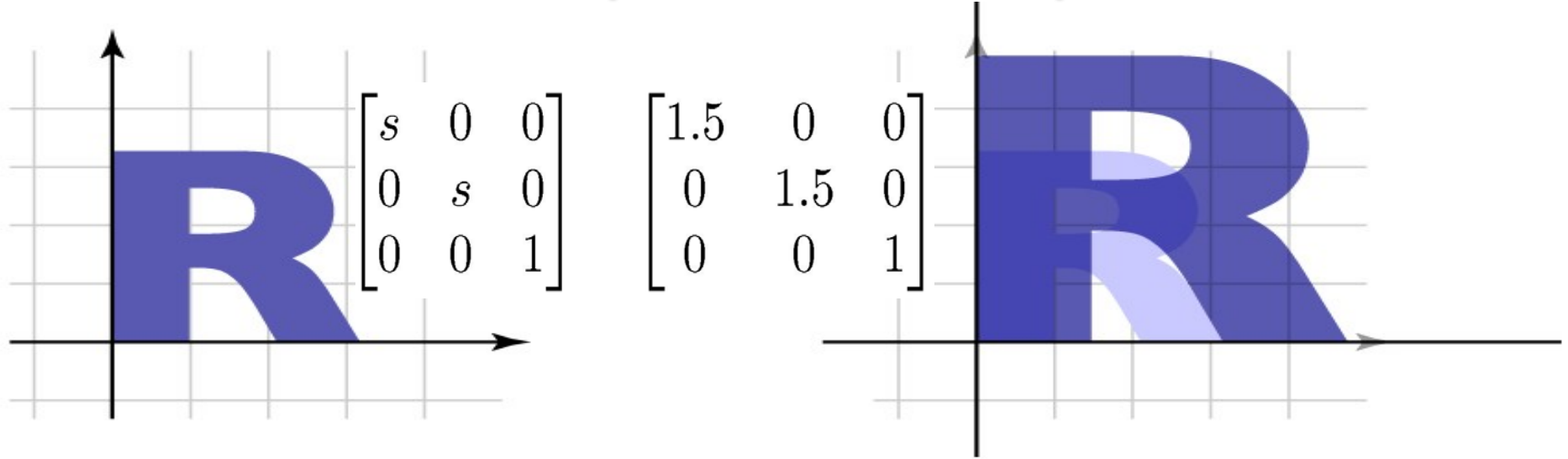
$(k > 1)$

$$\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$$



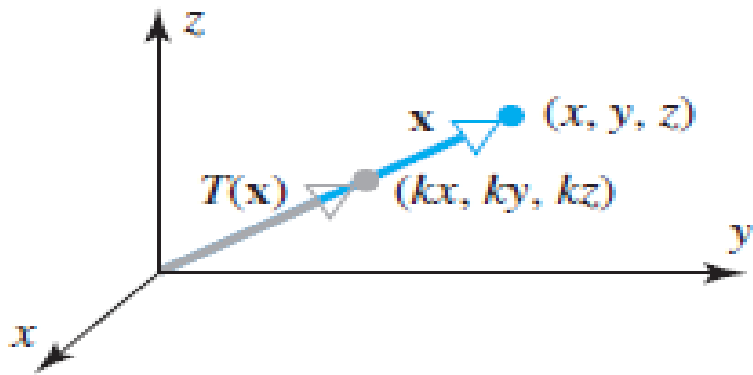
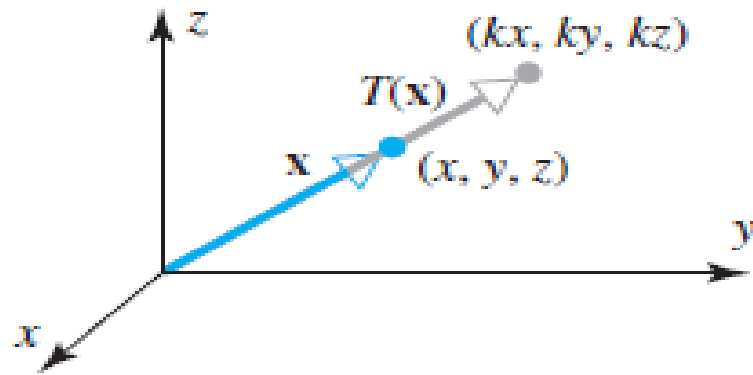
Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

Section 4.11 Geometry of Matrix Operators on $\mathbb{R}^2$



Section 4.9 Matrix Transformations from R^n to R^m

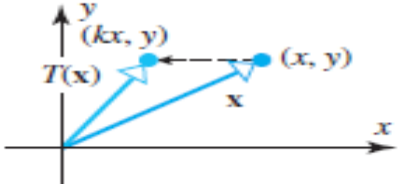
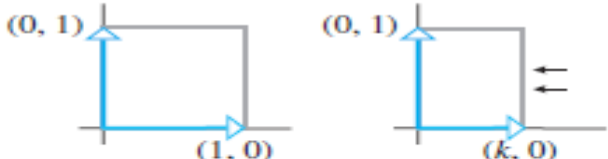
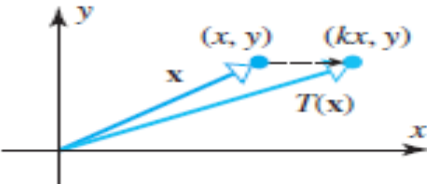
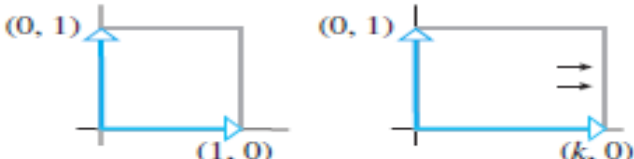
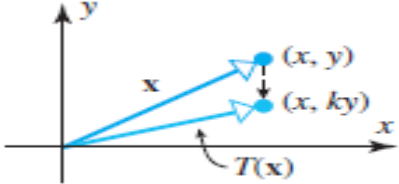
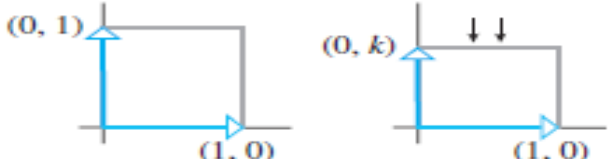
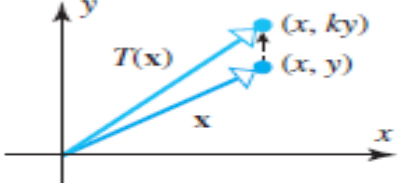
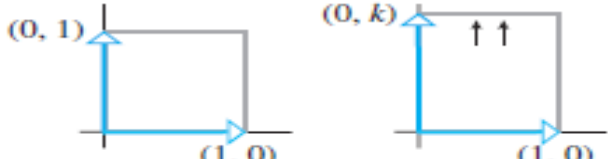
Table 8

Operator	Illustration $T(x, y, z) = (kx, ky, kz)$	Standard Matrix
Contraction with factor k on R^3 $(0 \leq k \leq 1)$		$\begin{bmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k \end{bmatrix}$
Dilation with factor k on R^3 $(k \geq 1)$		

Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

Expansion or Compression

Table 9

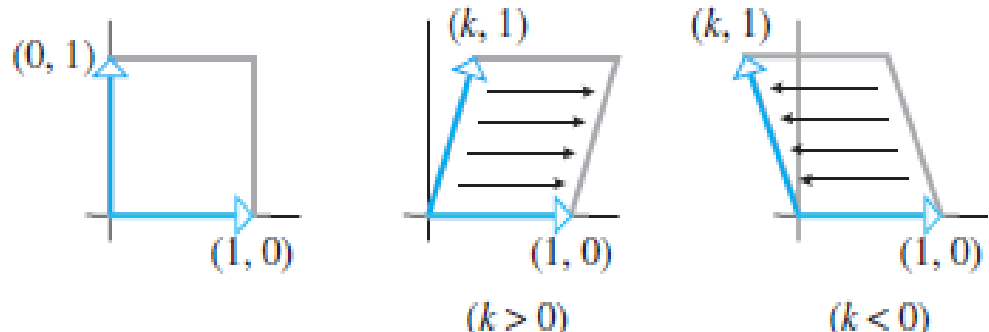
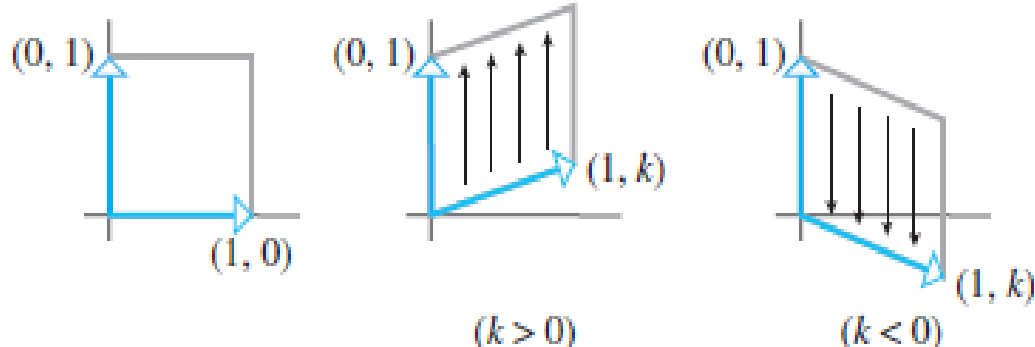
Operator	Illustration $T(x, y) = (kx, y)$	Effect on the Standard Basis	Standard Matrix
Compression of $\mathbb{R}^2$ in the x -direction with factor k ($0 \leq k < 1$)			$\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$
Expansion of $\mathbb{R}^2$ in the x -direction with factor k ($k > 1$)			
Operator	Illustration $T(x, y) = (x, ky)$	Effect on the Standard Basis	Standard Matrix
Compression of $\mathbb{R}^2$ in the y -direction with factor k ($0 \leq k < 1$)			$\begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$
Expansion of $\mathbb{R}^2$ in the y -direction with factor k ($k > 1$)			

Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

Section 4.11 Geometry of Matrix Operators on $\mathbb{R}^2$

Shear

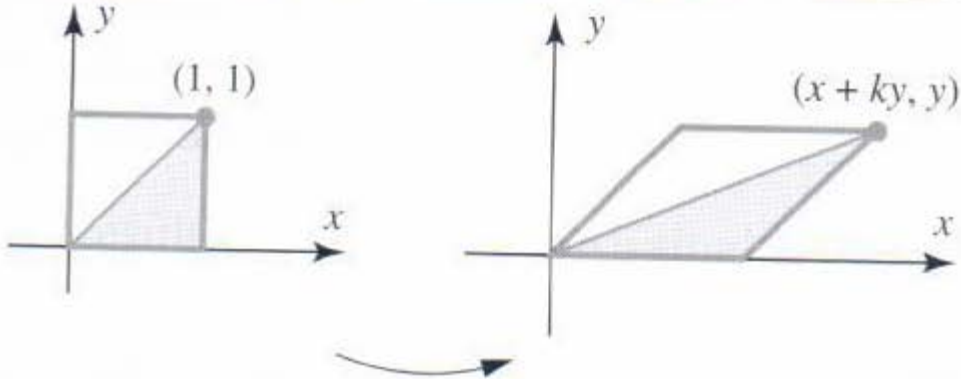
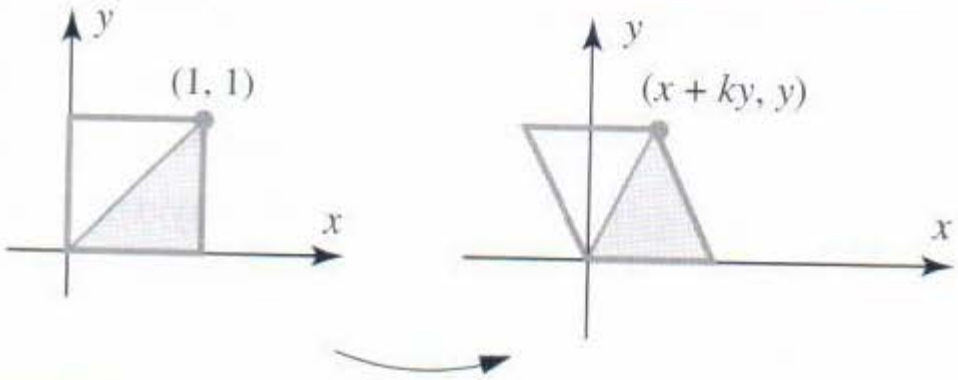
Table 10

Operator	Effect on the Standard Basis	Standard Matrix
Shear of $\mathbb{R}^2$ in the x-direction with factor k $T(x, y) = (x + ky, y)$		$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$
Shear of $\mathbb{R}^2$ in the y-direction with factor k $T(x, y) = (x, y + kx)$		$\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$

Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

Section 4.11 Geometry of Matrix Operators on $\mathbb{R}^2$

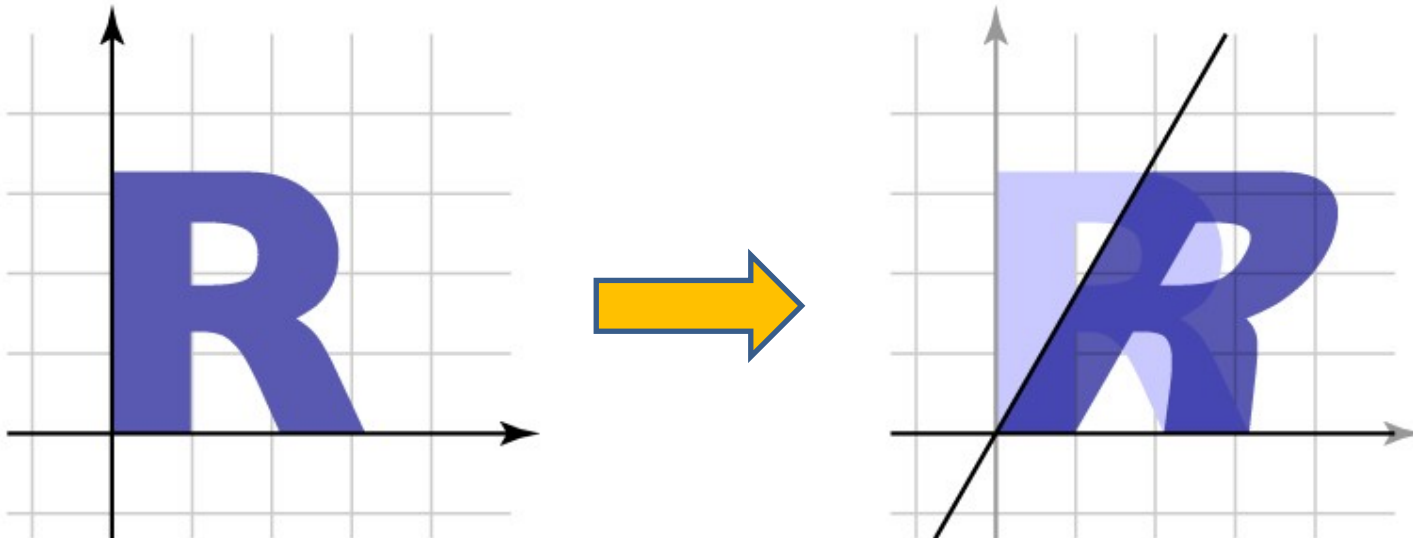
Shear

Shear in the x -direction with factor $k > 0$	$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$	
Shear in the x -direction with factor $k < 0$	$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$	

Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

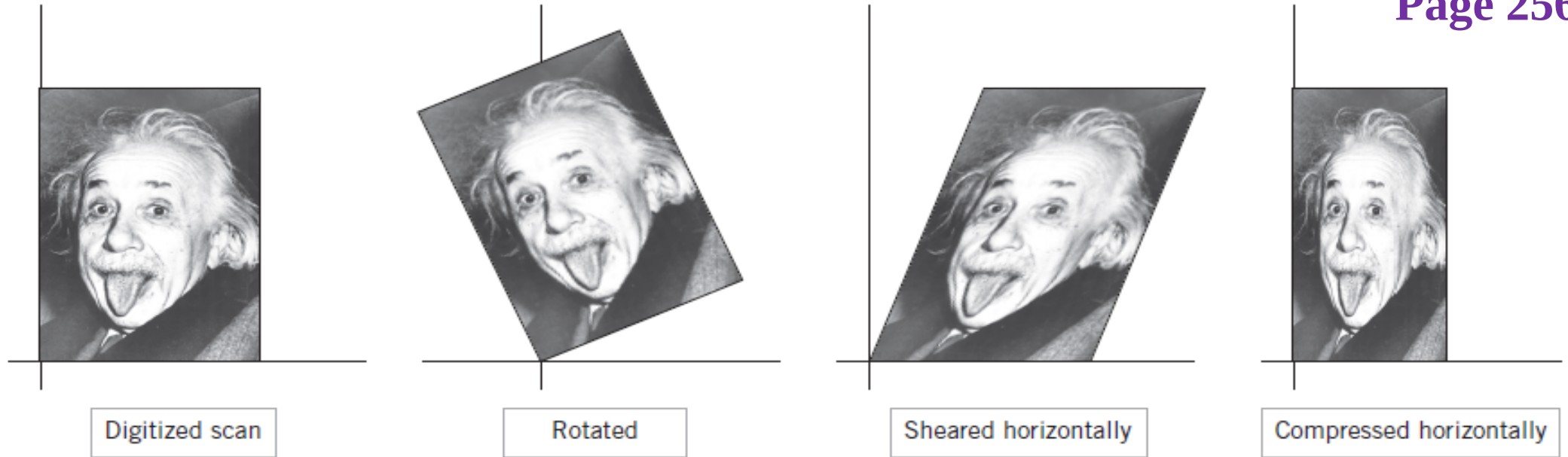
Shear

$$\begin{bmatrix} 1 & a & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0.5 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

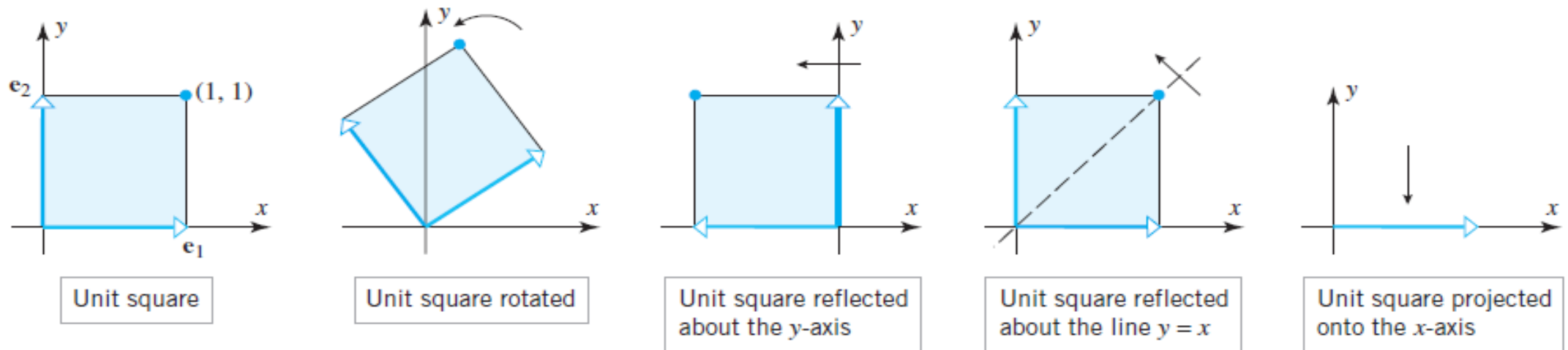


Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

Page 256



▲ Figure 4.11.1



Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

▶ EXAMPLE 1 A matrix Transformation from $\mathbb{R}^4$ to $\mathbb{R}^3$

The matrix transformation $T: \mathbb{R}^4 \rightarrow \mathbb{R}^3$ defined by the equations

$$w_1 = 2x_1 - 3x_2 + x_3 - 5x_4$$

$$w_2 = 4x_1 + x_2 - 2x_3 + x_4$$

$$w_3 = 5x_1 - x_2 + 4x_3$$

can be expressed in matrix form as

$$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} 2 & -3 & 1 & -5 \\ 4 & 1 & -2 & 1 \\ 5 & -1 & 4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \quad (8)$$

so the standard matrix for T is

$$A = \begin{bmatrix} 2 & -3 & 1 & -5 \\ 4 & 1 & -2 & 1 \\ 5 & -1 & 4 & 0 \end{bmatrix}$$

The image of a point $(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4)$ can be computed directly from the defining equations (7) or from (8) by matrix multiplication. For example, if

$$(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4) = (1, -3, 0, 2)$$

then substituting in (7) yields $w_1 = 1$, $w_2 = 3$, $w_3 = 8$ (verify), or alternatively from (8),

$$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} 2 & -3 & 1 & -5 \\ 4 & 1 & -2 & 1 \\ 5 & -1 & 4 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -3 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 8 \end{bmatrix}$$

Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

THEOREM 4.9.1 For every matrix A the matrix transformation $T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m$ has the following properties for all vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ and for every scalar k :

(a) $T_A(\mathbf{0}) = \mathbf{0}$

(b) $T_A(k\mathbf{u}) = kT_A(\mathbf{u})$ [Homogeneity property]

(c) $T_A(\mathbf{u} + \mathbf{v}) = T_A(\mathbf{u}) + T_A(\mathbf{v})$ [Additivity property]

(d) $T_A(\mathbf{u} - \mathbf{v}) = T_A(\mathbf{u}) - T_A(\mathbf{v})$

THEOREM 4.9.2 If $T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m$ are matrix transformations, and if $T_A(\mathbf{x}) = T_B(\mathbf{x})$ for every vector $\mathbf{x}$ in $\mathbb{R}^n$, then $A = B$

Homework

Sec. 4.9 Study this example

► **EXAMPLE 5** Some basic Matrix Operators R^2

In each part, describe the matrix operator corresponding to A , and show its effect on the unit square.

$$(a) \ A_1 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \quad (b) \ A_2 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \quad (c) \ A_3 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

Solution By comparing the forms of these matrices to those in Tables 7, 9, and 10, we see that the matrix A_1 corresponds to a shear in the x -direction by a factor 2, the matrix A_2 corresponds to a dilation with factor 2, and A_3 corresponds to an expansion in the x -direction with factor 2. The effects of these operators on the unit square are shown in Figure 4.9.6 .

► **EXAMPLE 5**

Some basic Matrix Operators R^2

$$(a) A_1 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \quad (b) A_2 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \quad (c) A_3 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

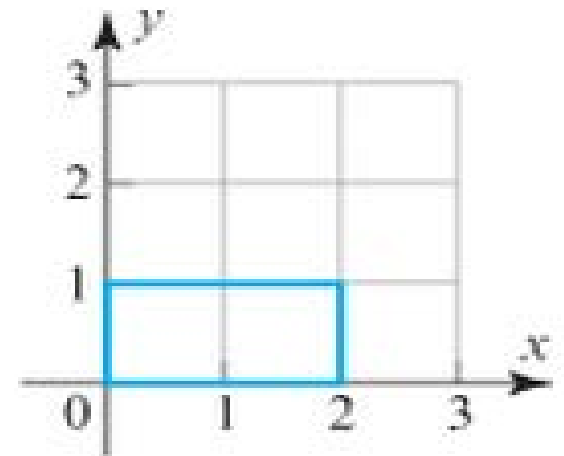
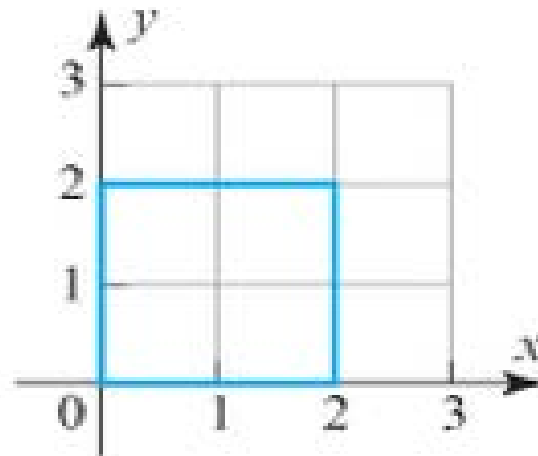
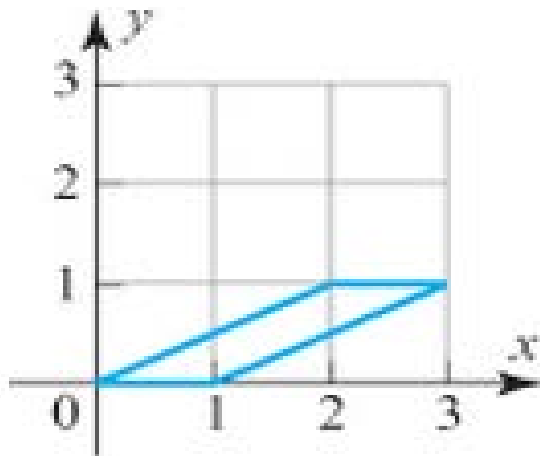


Figure 4.9.6